

# MATH403: Homework 6

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**18.** Let  $k$  be a divisor of  $n$ . To what familiar group is  $Z_n/\langle k \rangle$  isomorphic?

*Proof.* By definition of factor group,

$$Z_n/\langle k \rangle = \{a + \langle k \rangle \mid a \in Z_n\} = \{0 + \langle k \rangle, 1 + \langle k \rangle, \dots, (n-1) + \langle k \rangle\}$$

where recall that  $Z_n = \{0, 1, \dots, n-1\}$  and the subgroup  $\langle k \rangle = \{0, k, 2k, \dots, (m-1)k\}$  where  $m = \frac{n}{k}$ . However, observe the last term  $(n-1) + \langle k \rangle$ . Since  $k$  is a divisor of  $n$ , the multiples of  $k$  will cycle through the elements of  $Z_n$  every  $k$  steps. Thus, the cosets in the factor group are

$$Z_n/\langle k \rangle = \{0 + \langle k \rangle, \dots, (k-1) + \langle k \rangle\}$$

Thus, the factor group contains exactly  $k$  distinct cosets, which is the same order as  $Z_k$ . Thus, let us define the mapping  $\phi : Z_n/\langle k \rangle \rightarrow Z_k$  where for all cosets  $a + \langle k \rangle \in Z_n/\langle k \rangle$ ,  $\phi(a + \langle k \rangle) = a \bmod k$ . Note that this mapping is clearly 1-1 and onto because each element in  $Z_k$  corresponds to exactly one coset in  $Z_n/\langle k \rangle$ . Furthermore,  $\phi$  is operation preserving because  $\phi((a+b) + \langle k \rangle) = \phi(a + \langle k \rangle) + \phi(b + \langle k \rangle)$ , and thus  $Z_n/\langle k \rangle$  is isomorphic to  $Z_k$ .  $\square$

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- 24.** The group  $(Z_4 \oplus Z_{12}) / \langle (2, 2) \rangle$  is isomorphic to one of  $Z_8$ ,  $Z_4 \oplus Z_2$ , or  $Z_2 \oplus Z_2 \oplus Z_2$ . Determine which one by elimination.

*Proof.* Note that the group  $(Z_4 \oplus Z_{12}) / \langle (2, 2) \rangle$ , by the definition of factor groups, is

$$G/H = \{aH \mid a \in G\} \implies \{a\langle (2, 2) \rangle \mid a \in Z_4 \oplus Z_{12}\}$$

where  $|Z_4 \oplus Z_{12}| = 48$  and  $|\langle (2, 2) \rangle| = \text{lcm}(2, 6) = 6 \implies |(Z_4 \oplus Z_{12}) / \langle (2, 2) \rangle| = 8$ . Specifically,  $\langle (2, 2) \rangle = \{(0, 0), (2, 2), (0, 4), (2, 6), (0, 8), (2, 10)\}$ . Furthermore, the distinct cosets are the representatives  $a \in G$  is

$$a \in \{(0, 0), (1, 0), (2, 0), (3, 0), (0, 1), (0, 2), (0, 3), (0, 4)\}$$

At this point, none of the choices are eliminated because all of the groups have order 8; thus, let us analyze the structure of the quotient group. Note that the presence of elements of order 4 and order 2 suggest that the quotient group has structure similar to  $Z_4 \oplus Z_2$ .

The quotient group is not isomorphic to  $Z_8$  because the quotient group is not cyclic whereas  $Z_8$  can be generated by 1. The quotient group is not isomorphic to  $Z_2 \oplus Z_2 \oplus Z_2$  because every non-identity element in this group has order 2, and this does not correspond to the orders of the cosets in the quotient group.  $\square$

**32.** If  $N$  is a normal subgroup of  $G$  and  $|G/N| = m$ , show that  $x^m \in N$  for all  $x$  in  $G$ .

*Proof.* Since the order of the quotient group  $|G/N| = m \implies$  there are  $m$  distinct cosets, and we can represent them as

$$N, g_1N, g_2N, \dots, g_{m-1}N$$

If  $x = e \in G$ , then trivially  $x^m = e^m \in N$ . However, if  $x \in G$  and  $x \neq e$ , then  $x \in xN = g_iN$ , for some arbitrary coset representative denoted by  $i$ . Now consider raising both sides to the power of  $m$ ,

$$x^m \in (xN)^m = (g_iN)^m = g_i^m N$$

by the group operation of factor groups in **Theorem 9.2**. Now, to show that  $x^m \in N$ , we have to show that  $g_i^m = e$ . Recall that  $|G/N|$  is finite, and  $g_i \in G/N$ . Then by **Corollary 4**,

$$g_i^{|G/N|} = g_i^m = g_i^{|g_i|k} = e^k = e$$

for some  $k \in \mathbb{R}^+$ , and thus we have shown that  $x^m \in N$ , as desired.  $\square$

**38.** Prove that for every positive integer  $n$ ,  $Q/Z$  has an element of order  $n$ .

*Proof.* Intuitively,  $Z$  is a subset of  $Q$ . For  $a, b \in Z$ , in additive notation, note that  $a - b \in Z$ , so by the **One-Step Subgroup Test**,  $Z$  is a subgroup of  $Q$ . Now let us show that the subgroup  $Z$  of  $Q$  is normal. Recall **Theorem 9.1**, which states that a subgroup  $H$  of  $G$  is normal in  $G$  if and only if  $xHx^{-1} \subseteq H \forall x \in G$ . Let  $x = \frac{m}{n} \in Q$ , and observe that in additive notation,  $xHx^{-1} \implies x + H - x = \frac{m}{n} + H - \frac{m}{n} = H \implies xHx^{-1} \subseteq H$ . Therefore,  $Z$  is also normal in  $G$ . Now consider the factor group

$$Q/Z = \{a + Z \mid a \in Q\}$$

We want to show that there exists an element in the quotient group of order  $n$ , where  $n \in \mathbb{Z}^+$ . Consider, the coset  $\frac{1}{n} + Z$  given by the representative  $\frac{1}{n}$ . We will show that this element has order  $n$ .

$$n \left( \frac{1}{n} + Z \right) = 1 + Z = Z$$

the identity coset. Note that in  $Q/Z$ , two rational numbers  $a$  and  $b$  are considered equivalent if  $a - b \in Z$  because they belong to the same equivalence class. Furthermore,  $\frac{1}{n} * k$  for  $k < n$  is not the identity because  $\frac{k}{n}$  is not an integer, so  $n$  is the smallest number such that  $n(\frac{1}{n} + Z) = Z$ , so this element has order  $n$ , as desired.  $\square$

**60.** Without looking at inner automorphisms of  $D_n$ , determine the number of such automorphisms.

*Proof.* By **Theorem 9.6**, for any group  $G$ ,  $G/Z(G)$  is isomorphic to  $\text{Inn}(G)$ , so thus, it suffices to find  $|G/Z(G)| = |G|/|Z(G)|$ . When  $n$  is odd, the only elements that commutes with every other element is the identity. When  $n$  is even, both  $R_{180}$  and  $e$  commute with all other elements. Therefore, when  $n$  is odd, there are  $|G|/|Z(G)| = 2n$  inner automorphisms, and when  $n$  is even, there are  $n$  inner automorphisms.  $\square$

88. Explain why there is no group  $G$  with the property that  $G/Z(G)$  is isomorphic to the integers under addition.

*Proof.* Note that the group of integers under addition is an infinite, cyclic, Abelian group. We will show that there does not exist a group  $G$  such that  $G/Z(G)$  is isomorphic to the integers under addition. Recall two important properties of isomorphisms acting on groups: suppose  $G$  is isomorphic to  $G'$ , then  $G$  is Abelian if and only if  $G'$  is Abelian and  $G$  is cyclic if and only if  $G'$  is cyclic.

We will proceed with contradiction. Assume on the contrary that  $G/Z(G)$  is isomorphic to  $Z$  under addition. This implies that  $G/Z(G)$  is Abelian, but if  $G$  is non-Abelian, then the quotient group to its center cannot be Abelian, and so we have reached a contradiction because a non-Abelian group cannot be isomorphic to an Abelian group. If  $G$  is Abelian, then  $G/Z(G)$  would be of order 1 and isomorphic to the identity, which cannot be  $Z$ . Thus, there does not exist a group such that  $G/Z(G) \approx Z$  under addition.  $\square$

**90.** Let  $G$  be a group. Prove that  $\text{Inn}(G)$  is normal in  $\text{Aut}(G)$ .

*Proof.*

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